

EgoScore — a curation engine for EgoVerse

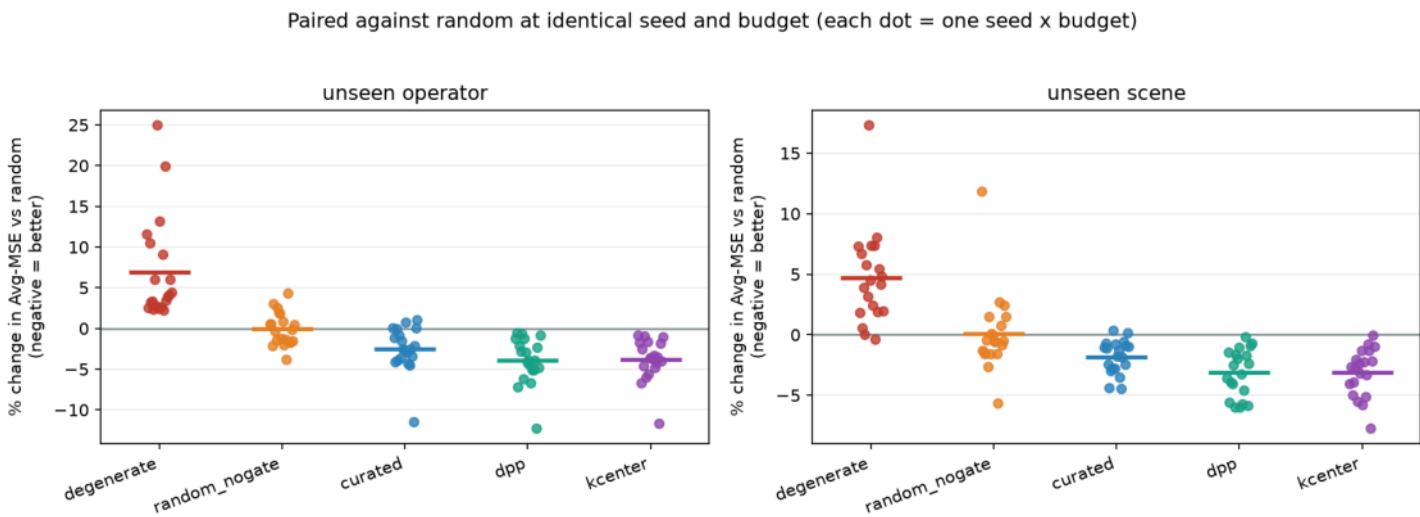
Track 1 · Which episodes are worth training on? · github.com/sahilmahendrakar/egoscore

We picked a quarter of the data and measured whether that quarter trained a better model than a quarter picked at random. It did.

RESULT

10 seeds × 2 budgets × 2 held-out axes = 40 head-to-head tests

SELECTOR	WINS	AVG-MSE	p
dpp — log-det diversity	40/40	-3.53%	2e-12
k-center	40/40	-3.45%	2e-12
facility location	33/40	-2.14%	2e-08
no quality gate	24/40	+0.02%	n.s.
degenerate — the control	1/40	+5.83%	5e-12



Not “delete 75% of your data for free” — training on everything still wins.
At a quarter of the budget, diversity selection recovers 49% (unseen operator) / 41% (unseen scene) of the gap to using everything.

Why believe a 3.5% margin

We added a method designed to lose: the same budget concentrated into ~20 operator×scene groups instead of ~95.

It loses 1/40 at +5.83%.

The EgoVerse paper already showed demonstrator and scene variety help. Our setup reproduces that, so it can detect a real effect — and the smaller numbers are measurements, not luck.

The gate, priced

rl2 drops 49 of 572 (8.6%) across four reasons. microagi — the largest fold_clothes slice — drops 13.9%, all of them recordings running past 3× that lab’s median length.

Matched on episode count, dropping them looks harmful (+1.68%). That’s an artefact: they carry 60× the frames.

Matched on frames, the gate wins 10/10 seeds at -4.66%. They were never better data, just more.

Three things in the data

Zarr arrays are zero-padded to a chunk boundary. Read them raw and the tail looks exactly like a frozen tracker.

Video is 84.6 GB for this slice; poses are 2.3 GB. Every signal is pose-only on purpose — curation must cost less than the training it saves.

An “episode” is not a common unit: 93 s in rl2, 11 s in microagi. Budget in seconds, not episodes.

Limitations

Avg-MSE is a proxy, not robot success — the EgoVerse authors say so and use it anyway.

The scoring model never sees video, which follows from the 84.6 GB figure.

One task, one lab. No claim of transfer.

Facility location was our pick going in and it lost to log-det and k-center. Reported, not quietly reordered.